

- **Features:**
 - EC2:** Virtual servers.
 - S3:** Scalable storage.
- **Google Cloud Platform (GCP):**
 - **Services:** Machine learning, data analytics, cloud storage.
 - **Features:**
 - **Compute Engine:** Virtual machines.
 - **Cloud Storage:** Unified object storage.
- **Microsoft Azure:**
 - **Services:** Cloud computing, storage, databases, AI.
 - **Features:**
 - **Virtual Machines:** Windows and Linux VMs.
 - **Azure SQL Database:** Managed relational database service.

How to use cloud storage effectively

Organising files on Google Drive and OneDrive

- **Folder Structure:**
 - **Create Folders:** Organise by project, date, or file type.
 - **Use Naming Conventions:** Consistent naming for easy retrieval.
- **Search Functionality:**
 - **Metadata Tags:** Add descriptions or tags to files.
 - **Advanced Search:** Use filters for file type, owner, or date modified.

Sharing and collaborating via the cloud

- **Setting Permissions:**
 - **View Only:** Recipients can view but not edit.
 - **Edit Access:** Allow collaborators to make changes.
 - **Comment Access:** Enable suggestions without altering the original content.
- **Collaboration Tools:**
 - **Real-Time Editing:** Multiple users can work on the same document simultaneously.
 - **Version History:** Track changes and revert to previous versions if necessary.

Data backup and recovery

Regular backups protect your data from loss due to hardware failure, accidental deletion, or cyber threats.

How to setup automated backups

- **Choose a Backup Solution:**
 - **Cloud-Based:** Services like Backblaze or Carbonite.
 - **External Drives:** Use software like Time Machine (Mac) or File History (Windows).
- **Schedule Backups:**
 - **Frequency:** Daily or weekly, depending on data changes.
 - **Retention Policy:** How long backups are kept before deletion.

Steps for setting up automated backups on Windows:

1. **Connect External Drive:** Ensure it's recognized by the system.
2. **Go to Settings:** Navigate to Update & Security > Backup.
3. **Add a Drive:** Select your external drive.
4. **Configure:** Choose folders to back up and schedule.

Choosing the right backup schedule

- **Critical Data:** Backup more frequently.
- **Large Files:** Schedule during off-peak hours to avoid slowing down your system.
- **Test Restores:** Periodically verify that backups can be successfully restored.

5) Gaining cybersecurity awareness skills

Cybersecurity is essential for protecting personal and professional data. Developing awareness and basic skills can prevent security breaches.

Learning basic cybersecurity skills

Importance of security Awareness in digital spaces

- **Protect Personal Information:** Safeguard against identity theft.
- **Prevent Financial Loss:** Avoid scams and fraudulent transactions.
- **Maintain Privacy:** Control over who accesses your data.

Recognising and avoiding online scams

Identifying common tactics: Phishing, social engineering

- **Phishing:**
 - **Emails or Messages:** Appear to be from legitimate sources.
 - **Objective:** Trick users into providing personal information.
 - **Red Flags:**
 - Urgent language.
 - Suspicious links or attachments.
 - Unusual sender addresses.
- **Social Engineering:**
 - **Manipulation:** Exploiting human psychology to gain access.
 - **Examples:** Impersonation, baiting with fake offers.

Resources for continuous cyber awareness

- **Online Courses:**
 - **Cybrary:** Free cybersecurity training.
 - **Coursera:** Courses on cybersecurity fundamentals.
- **Newsletters and Blogs:**
 - **Krebs on Security:** Updates on security news and threats.
 - **The Hacker News:** Latest cybersecurity incidents and best practices.
- **Government Resources:**
 - **US-CERT:** Alerts and tips from the U.S. Department of Homeland Security.
 - **Stay Safe Online:** Resources from the NSCA. 